INTRODUCTION TO DEEP LEARNING

Course number: 00240332

Lecture 8: Applications of CNN in Computer Vision

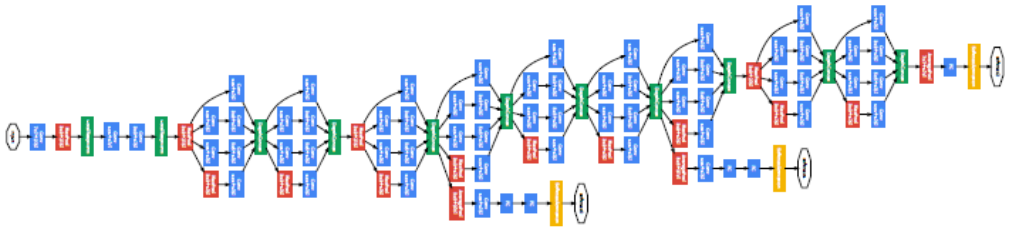
Xiaolin Hu (胡晓林)

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Tsinghua University

Last lecture review

I. Popular models



AlexNet

ResNet

GoogLeNet

VGGNet

DenseNet

II. Lightweight models

- Depthwise separable convolution
- Group convolution

MobileNet

ShuffleNet

ResNeXt

III. Compression and quantization*

- Model compression
- Model quantization

Network trimming

BinaryConnect

BinaryNet

Discussion

- Form groups of 2-3 and every group prepares a 5-minute presentation with slides for one of the following papers
  - Hu, Shen, Sun, “Squeeze-and-Excitation Networks, ” CVPR 2018
  - Li, Wang, Hu, Yang, “Selective Kernel Networks,” CVPR 2019

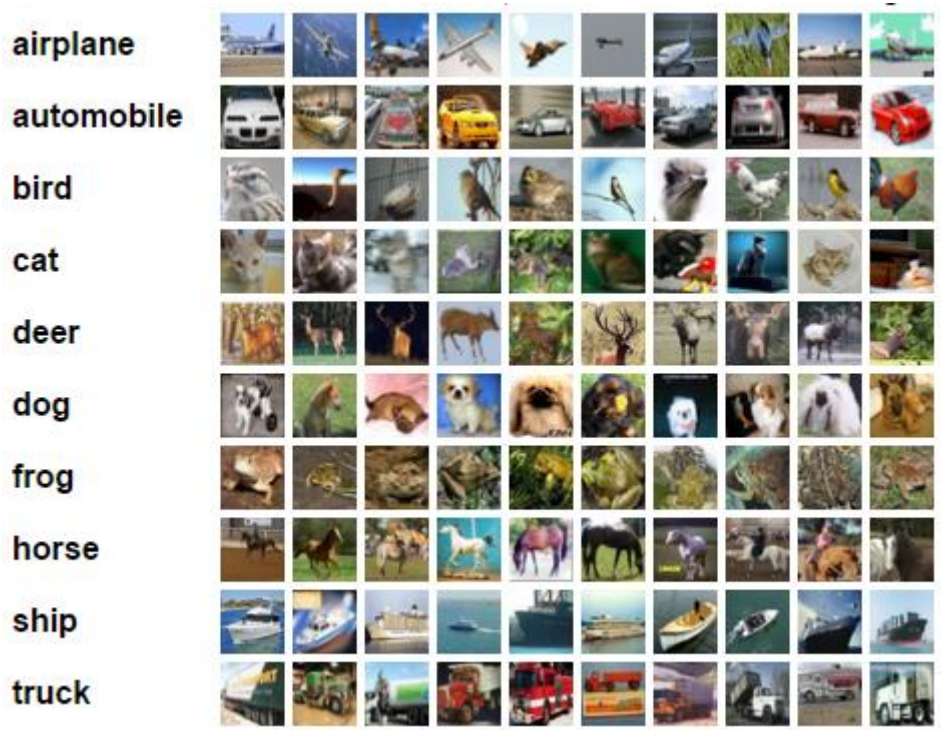
Outline

I. Image classification
II. Object detection
III. Image segmentation
IV. Summary

General image classification

CIFAR-10 & CIFAR100

- 50,000 training images and 10,000 test images
- 32x32 colour images



ImageNet competition (ILSVRC)

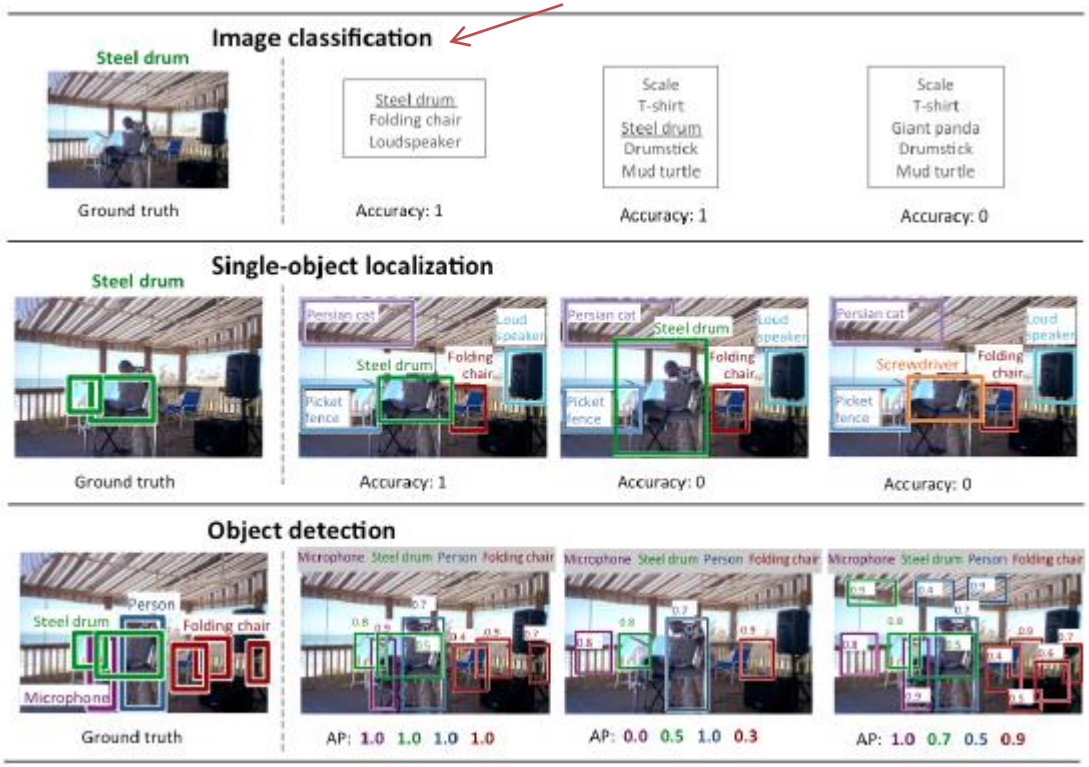
1M training images

Tasks

2010-

2011-

2013-



Top-1
Top-5 (preferred)

Two human expert: 5.1%, 12%

The first column shows the ground truth labeling on an example image, and the next three show three sample outputs with the corresponding evaluation score.

Russakovsky, et al., 2014

Specific image classification









Face verification





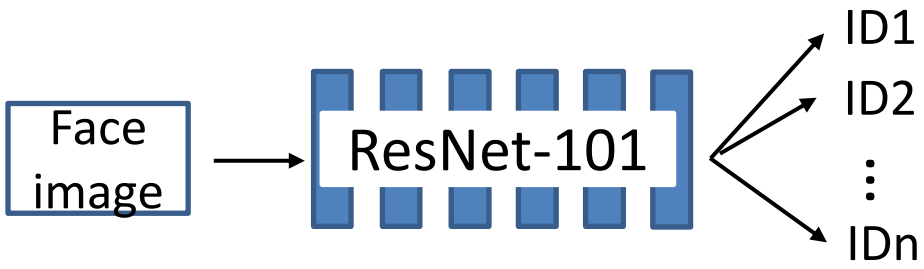


Model	Accuracy (%)
DeepFace (2014)	97.25
DeepID (2014)	97.45
DeepID2 (2014)	99.15
DeepID2+ (2014)	99.47
DeepID3 (2014)	99.53
FaceNet (2015)	99.63

- The task is actually different from image classification

Multi-class classification?

- Can you use a multi-class neural network-based classifier for this task and how?



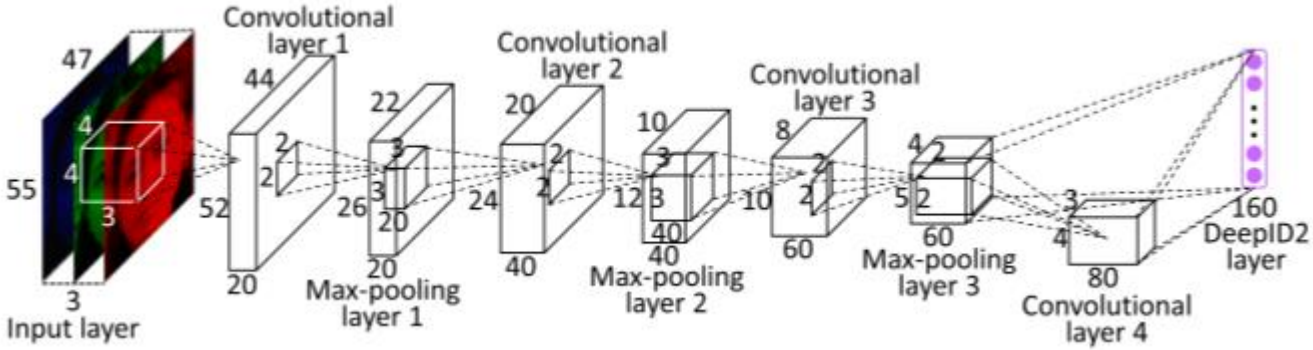
Problematic!

- ➢ Too many classes
  - Calculating the normalization term in softmax is expensive
  - Every softmax output is very small
- ➢ Too few samples in each class

What we really want is discrimination!

DeepID2

Sun, Wang, Tang, NeurIPS 2014



- Contrastive loss



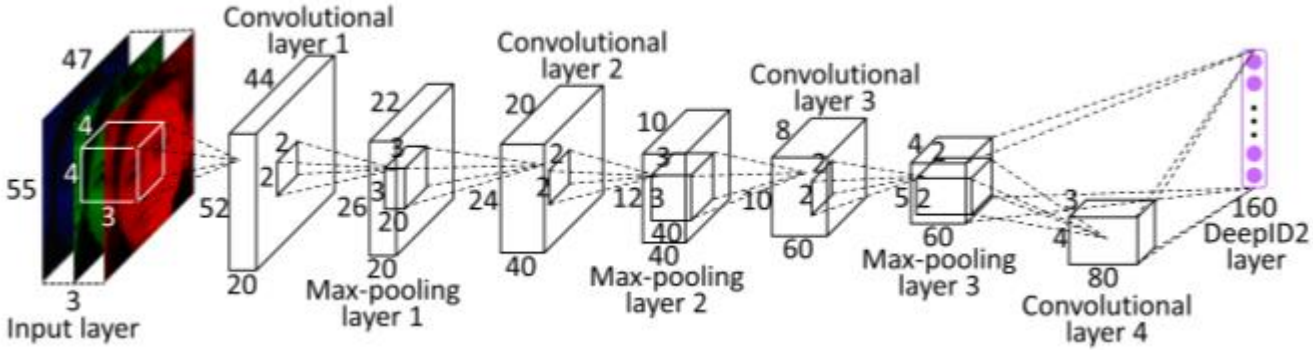
Goal:

$\left|\left|f(x_i) - f(x_j)\right|\right|_2^2$ is small as possible if i, j belong to the same identity;

$\left|\left|f(x_i) - f(x_j)\right|\right|_2^2 > \alpha$ otherwise, where $\alpha > 0$

DeepID2

Sun, Wang, Tang, NeurIPS 2014



- Contrastive loss



Loss:

$$L(x_i, x_j, y_{ij}, \theta) = \begin{cases} \frac{1}{2}\|f(x_i) - f(x_j)\|_2^2, & \text{if } y_{ij} = 1 \\ \frac{1}{2}\max\{0, \alpha - \|f(x_i) - f(x_j)\|_2^2\}, & \text{else.} \end{cases}$$

label



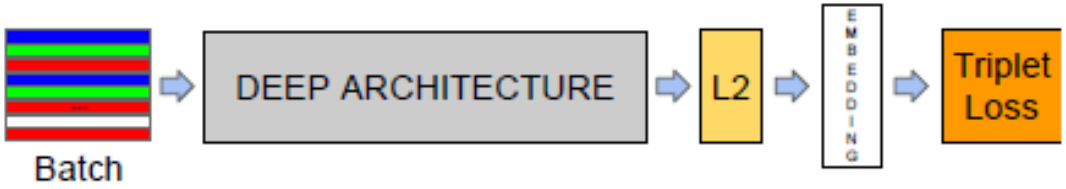
汤晓鸥 (1968-2023)

香港中文大学教授
商汤科技创始人

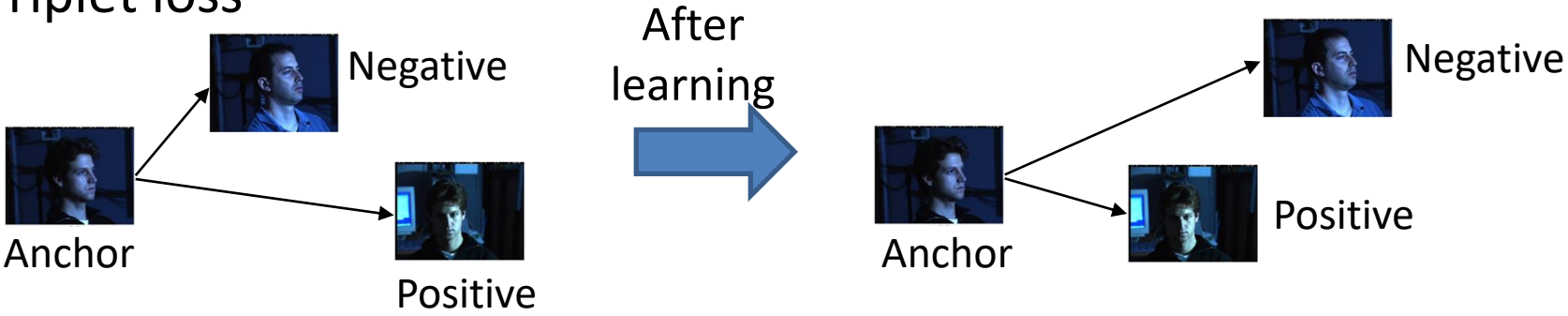


FaceNet

Schroff et al., CVPR 2015



- Triplet loss



Goal: $\left|\left|f(x_i^a) - f(x_i^p)\right|\right|_2^2 + \alpha < \left|\left|f(x_i^a) - f(x_i^n)\right|\right|_2^2$ where $\alpha > 0$

Loss: $\sum_i^N \left[\|f(x_i^a) - f(x_i^p)\|_2^2 - \|f(x_i^a) - f(x_i^n)\|_2^2 + \alpha\right]_+$

It indicates at least 3 samples are needed in a minibatch

Any problem with this?

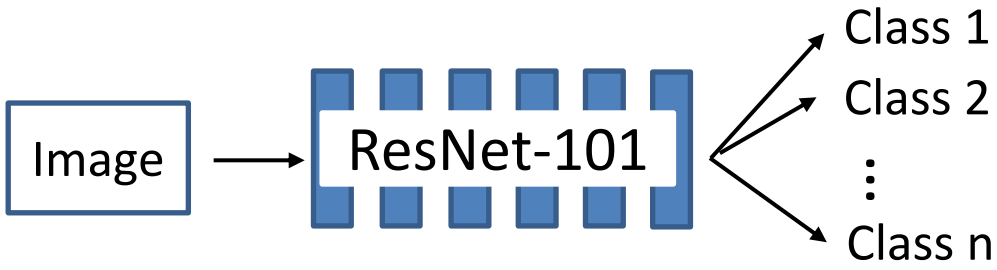
Other losses for face verification*

- SphereFace:
  - Deep Hypersphere Embedding for Face Recognition, CVPR 2017
- NormFace:
  - L2 Hypersphere Embedding for Face Verification, ACM MM 2017
- ArcFace:
  - Additive Angular Margin Loss for Deep Face Recognition, CVPR 2019

Summary of Part I

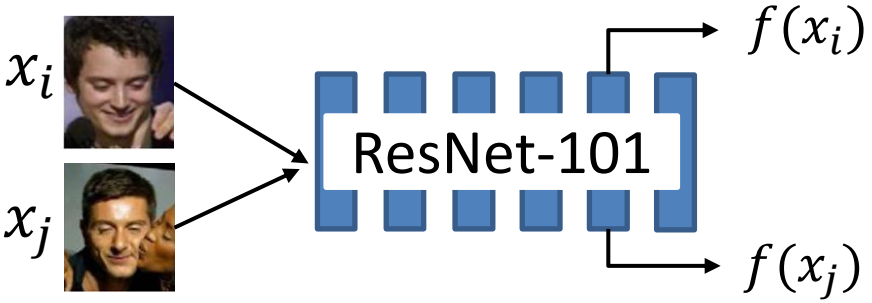
Multi-class classification

Q: Which class does this image belong to?

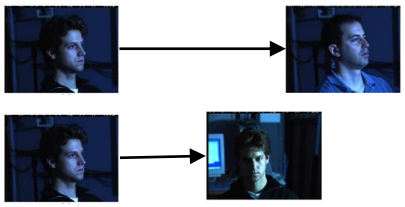


Verification

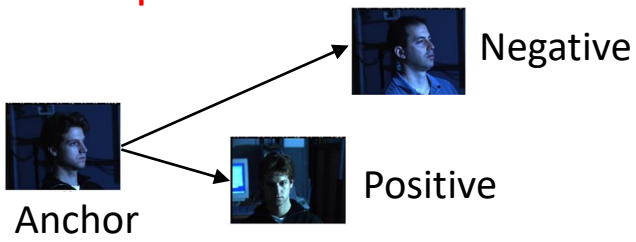
Q: Are the two images for the same person?



Contrastive loss



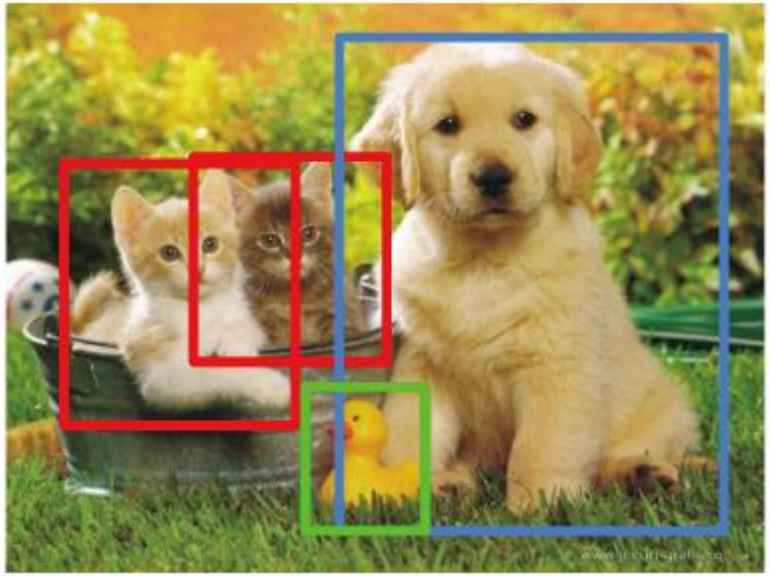
Triplet loss



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Task



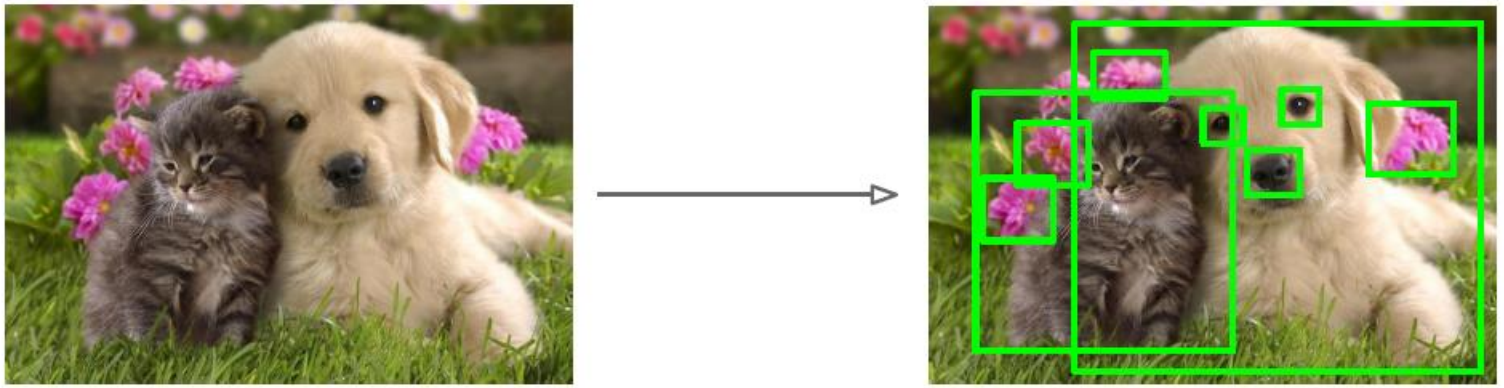
DOG, (x, y, w, h)
CAT, (x, y, w, h)
CAT, (x, y, w, h)
DUCK (x, y, w, h)

How would you do this?

A simple idea: find regions that contain objects, then use a classifier to do object recognition

Region proposals

- Find “blobby” image regions that are likely to contain objects
  - Such regions are called *region proposals*, or *region of interest (RoI)*



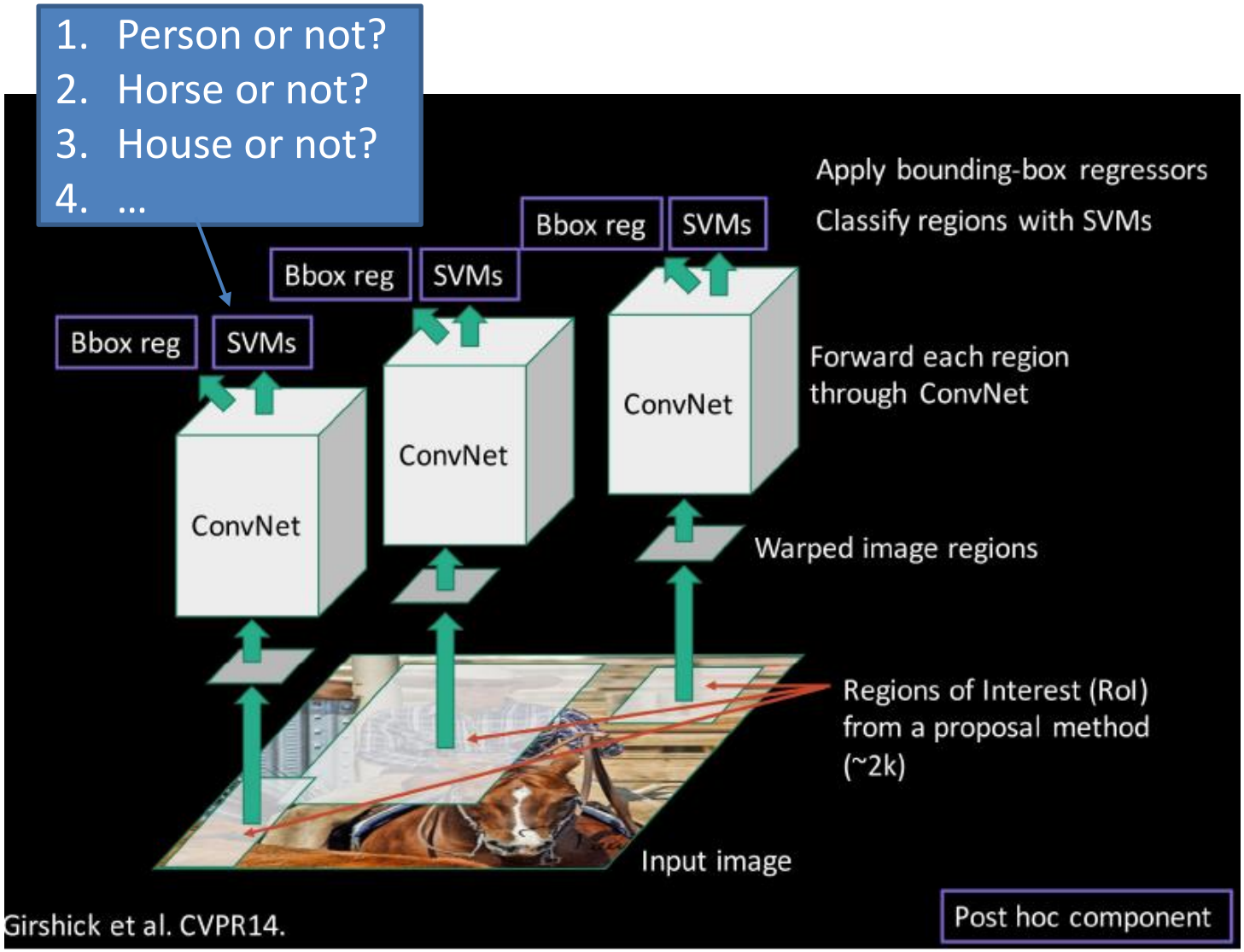
Modified from: Fei-Fei Li & Andrej Karpathy & Justin Johnson

Region proposal: many choices

Method	Approach	Outputs Segments	Outputs Score	Control #proposals	Time (sec.)	Repea-tability	Recall Results	Detection Results
Bing [18]	Window scoring		✓	✓	0.2	⋆⋆⋆	⋆	·
CPMC [19]	Grouping	✓	✓	✓	250	-	⋆⋆	⋆
EdgeBoxes [20]	Window scoring		✓	✓	0.3	⋆⋆	⋆⋆⋆	⋆⋆⋆
Endres [21]	Grouping	✓	✓	✓	100	-	⋆⋆⋆	⋆⋆
Geodesic [22]	Grouping	✓		✓	1	⋆	⋆⋆⋆	⋆⋆
MCG [23]	Grouping	✓	✓	✓	30	⋆	⋆⋆⋆	⋆⋆⋆
Objectness [24]	Window scoring		✓	✓	3	·	⋆	·
Rahtu [25]	Window scoring		✓	✓	3	·	·	⋆
RandomizedPrim's [26]	Grouping	✓		✓	1	⋆	⋆	⋆⋆
Rantalankila [27]	Grouping	✓		✓	10	⋆⋆	·	⋆⋆
Rigor [28]	Grouping	✓		✓	10	⋆	⋆⋆	⋆⋆
SelectiveSearch [29]	Grouping	✓	✓	✓	10	⋆⋆	⋆⋆⋆	⋆⋆⋆
Gaussian				✓	0	·	·	⋆
SlidingWindow				✓	0	⋆⋆⋆	·	·
Superpixels		✓			1	⋆	·	·
Uniform				✓	0	·	·	·

Hosang et al, PAMI 2015

R-CNN



Slide credit: Ross Girschick

Steps

1. Train (or download) a classification model for ImageNet (AlexNet)
2. Fine-tune model for detection
3. Extract features from each region
4. Train one binary SVM per class to classify region features
5. For each class, train a linear regression model to map from cached features to offsets to GT boxes to make up for “slightly wrong” proposals



Slide credit: Fei-Fei Li & Andrej Karpathy & Justin Johnson

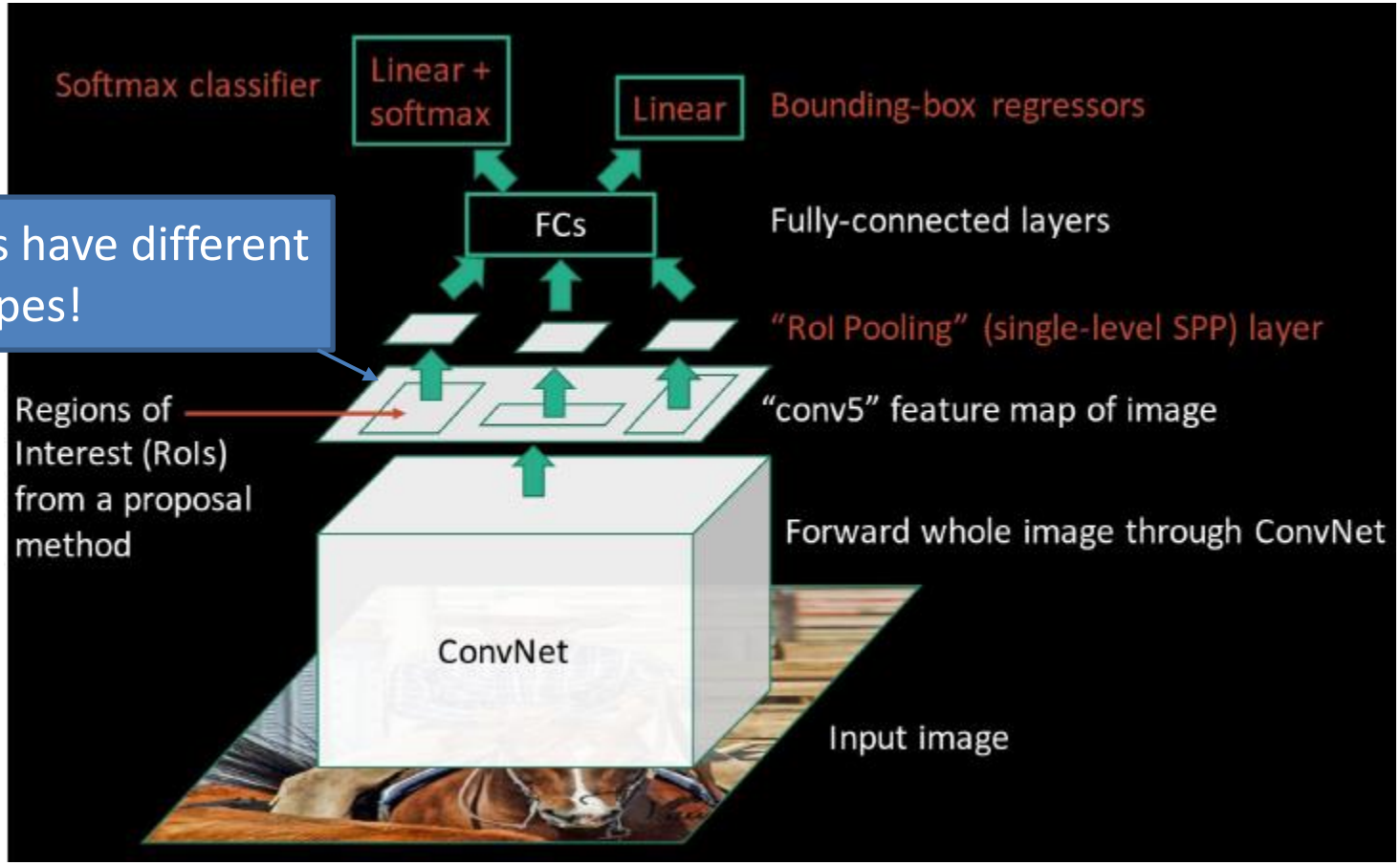
Any problem with R-CNN?

- Slow at test-time:
  - Need to run full forward pass of CNN for each region proposal
- SVMs and regressors are post-hoc:
  - CNN features not updated in response to SVMs and regressors

Do you have a better idea?

Let's run one forward pass of CNN on the entire image and map every ROI to the feature maps!

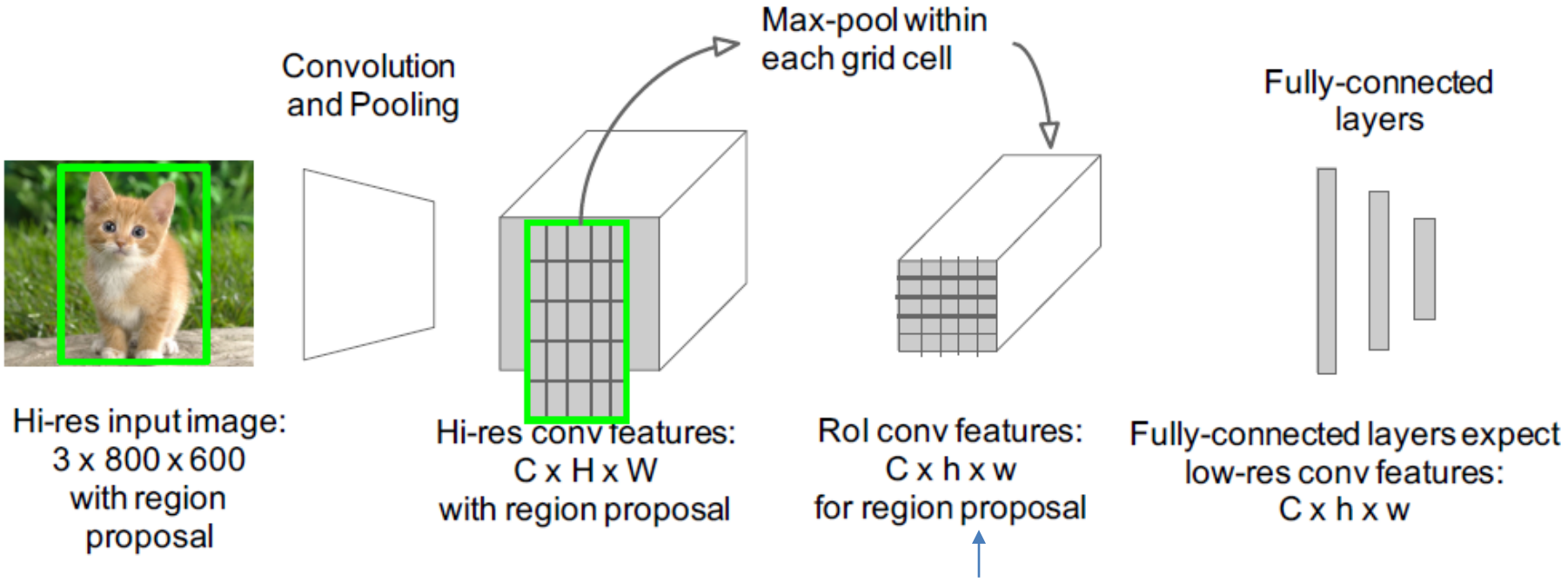
Fast R-CNN



Girshick, ICCV 2015

Slide credit: Ross Girschick

Region of interest pooling



Slide credit: Fei-Fei Li & Andrej Karpathy & Justin Johnson

Fast R-CNN Results

	R-CNN	Fast R-CNN
Training Time:	84 hours	**9.5 hours**
(Speedup)	1x	**8.8x**
Test time per image	47 seconds	**0.32 seconds**
(Speedup)	1x	**146x**
mAP (VOC 2007)	66.0	**66.9**

Faster!

FASTER!

Better!

Using VGG-16 CNN on Pascal VOC 2007 dataset

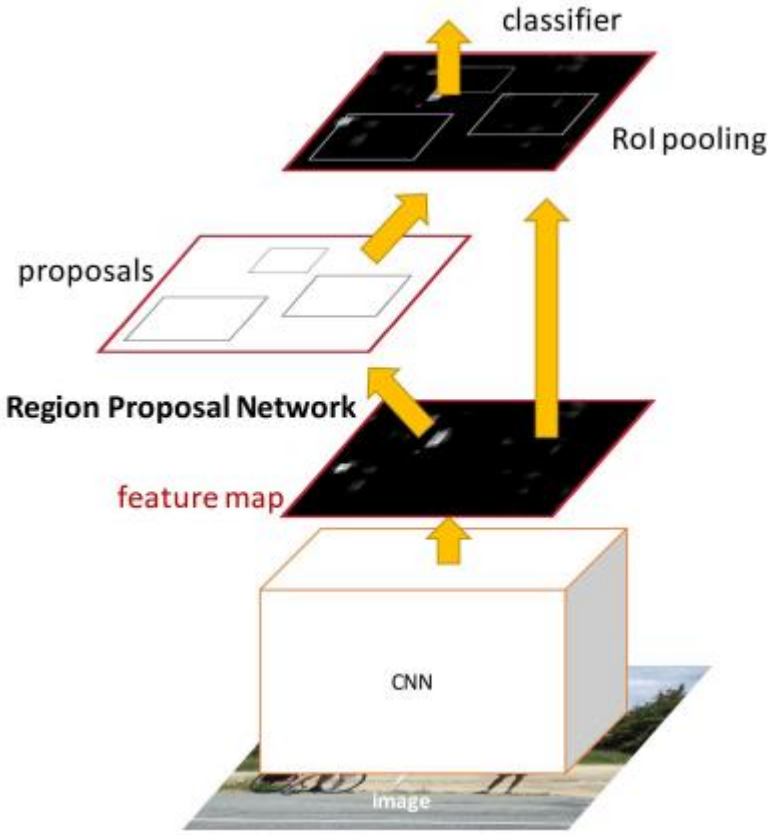
Slide credit: Fei-Fei Li & Andrej Karpathy & Justin Johnson

Question

Is there any problem with Fast R-CNN?

Faster R-CNN

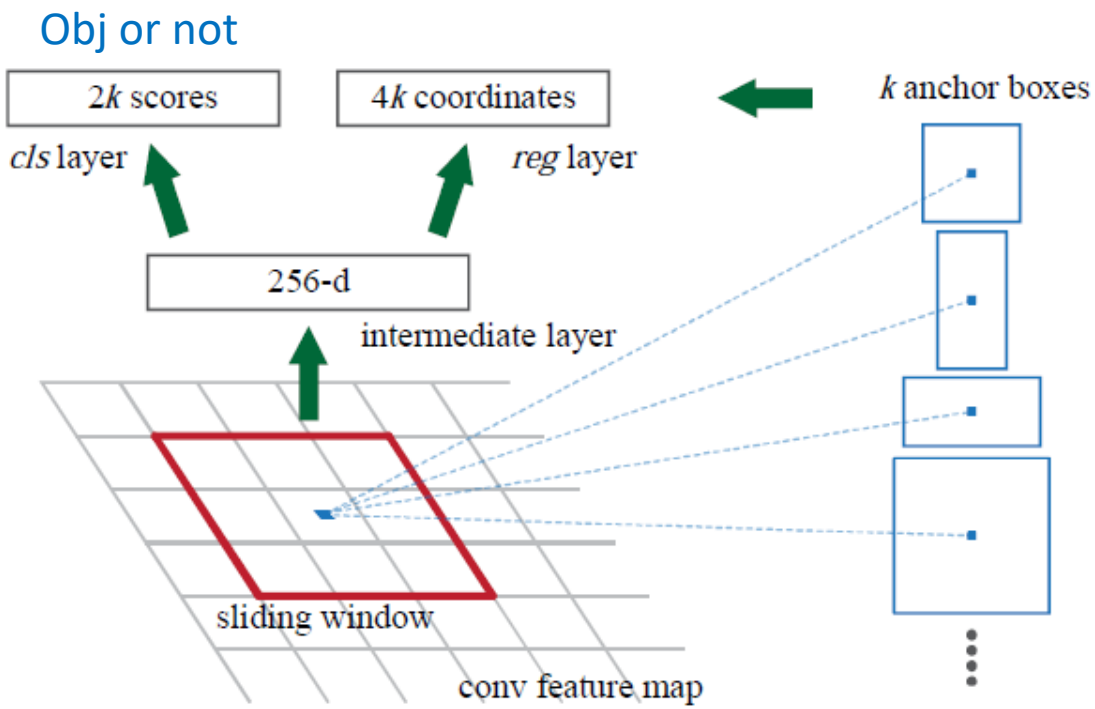
- Insert a Region Proposal Network (RPN) after the last convolutional layer
- RPN trained to produce region proposals directly; no need for external region proposals!
- After RPN, use RoI Pooling and an upstream classifier and bbox regressor just like Fast R-CNN



Ren et al., NIPS 2015

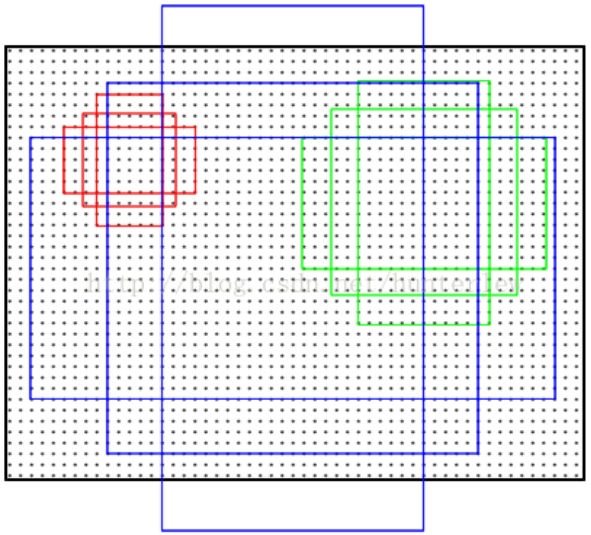
Slide credit: Ross Girschick

Region proposal network



Ren et al., NIPS 2015

At each location, k=9 boxes are generated in the input image



http://blog.csdn.NET/shenxiaolu1984/article/details/51152614

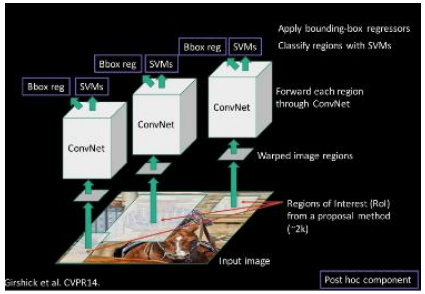
Faster R-CNN results

Table 2: Detection results on **PASCAL VOC 2007 test set**. The detector is Fast R-CNN and VGG-16. Training data: "07": VOC 2007 trainval, "07+12": union set of VOC 2007 trainval and VOC 2012 trainval. For RPN, the train-time proposals for Fast R-CNN are 2k. †: this was reported in [5]; using the repository provided by this paper, this number is higher (68.0±0.3 in six runs).

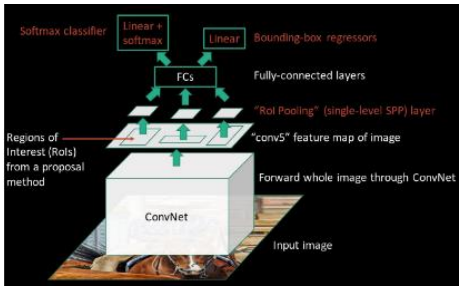
	method	# proposals	data	mAP (%)	time (ms)
Fast R-CNN	SS	2k	07	66.9†	1830
	SS	2k	07+12	70.0	1830
	RPN+VGG, unshared	300	07	68.5	342
	RPN+VGG, shared	300	07	69.9	**198**
	RPN+VGG, shared	300	07+12	**73.2**	**198**

Summary of Part II

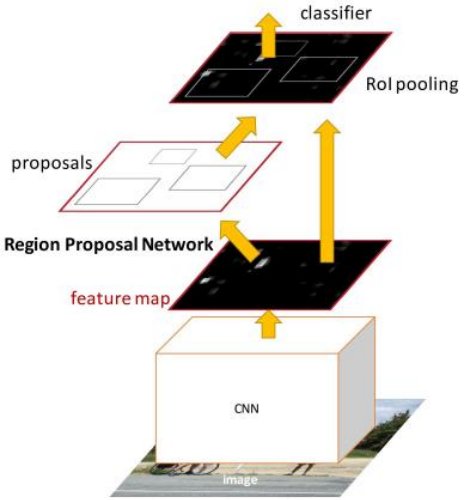
R-CNN



Fast R-CNN



Faster R-CNN



单选题 1分

Setting

How many samples at least are needed in a minibatch for minimizing the contrastive loss?

A 1

B 2

C 3

D 4

Submit

单选题 1分

Setting

For training the FaceNet, a triplet loss is used. After training, the distance between the features of an anchor sample and a positive sample should be

A reduced

B increased

C the same

Submit

多选题 1分

Setting

Which of the following generate RoIs using external models?

A R-CNN

B Fast R-CNN

C Faster R-CNN

Submit

多选题 1分

Setting

Faster R-CNN uses an RPN to generate proposals. What predictions does RPN make at every location?

A Coordinates of bounding boxes

B Whether the bounding boxes contain objects or not

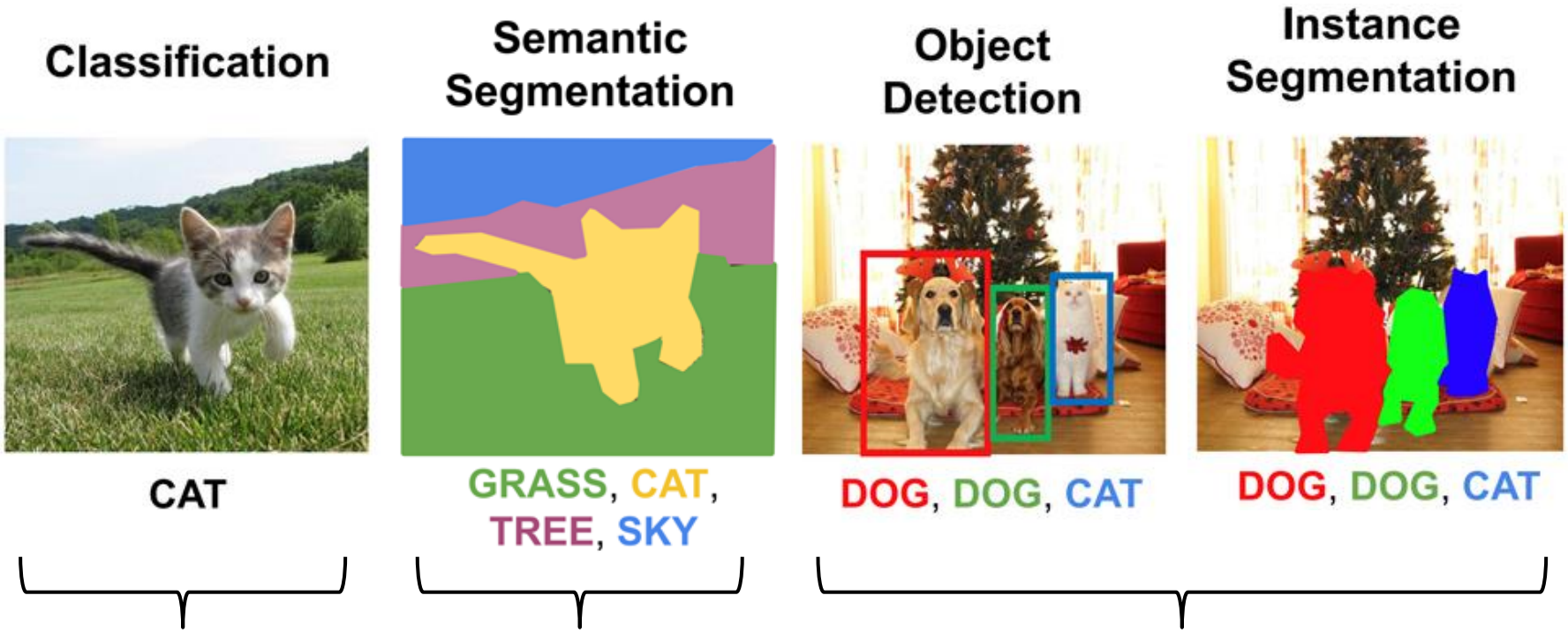
C Class labels of the objects

Submit

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I. Image classification
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Relation with other vision tasks



Semantic segmentation



CAT



GRASS, CAT,
TREE, SKY

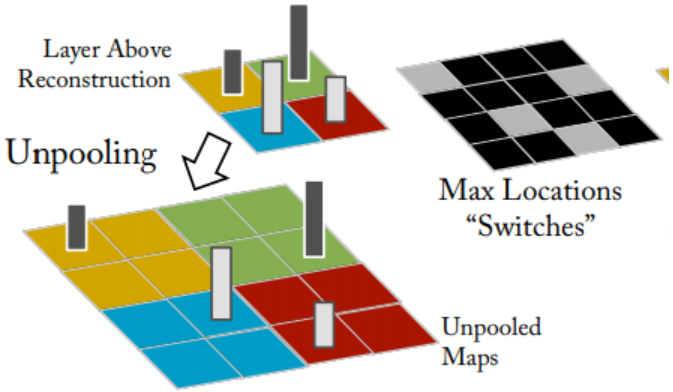
Input and output have the same spatial size



Usually, the size of feature maps is reduced in a network!

How to enlarge the feature maps

- Most CNNs need to enlarge the feature maps in certain layers. How would you do that?
- Upsampling (resampling and interpolation)
  - Take an input image, rescale it to the desired size and then calculate the pixel values at each point using a interpolation method such as bilinear interpolation
- Unpooling (reverse max pooling)
  - Record the locations of the maxima within each pooling region in a set of switch variables. Then place the reconstructions from the layer above into appropriate locations



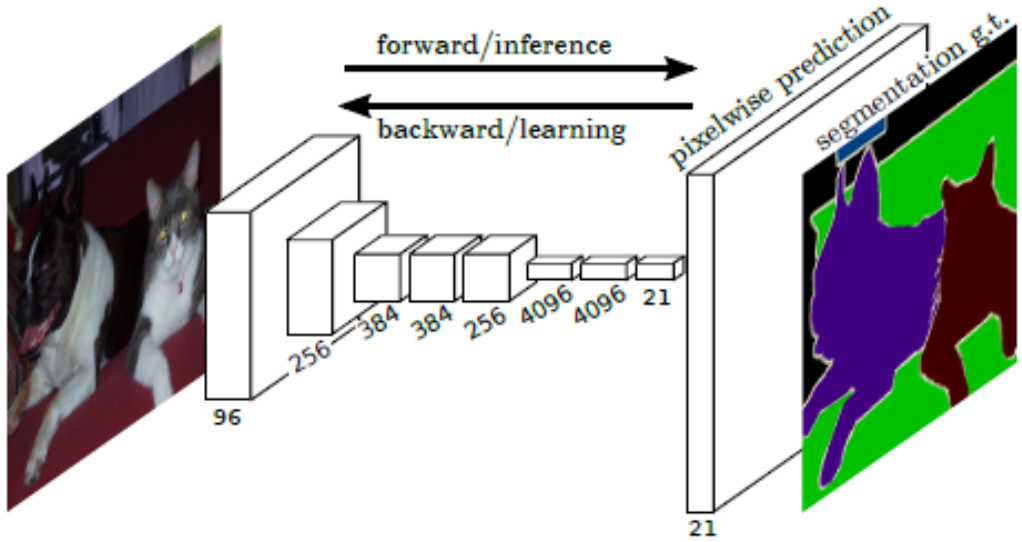
Zeiler, Fergus, 2013

- Transposed convolution (wrongly called *deconvolution*)
  http://deeplearning.net/software/theano/tutorial/conv_arithmetic.html

Fully Convolutional Networks

Long et al., CVPR 2015

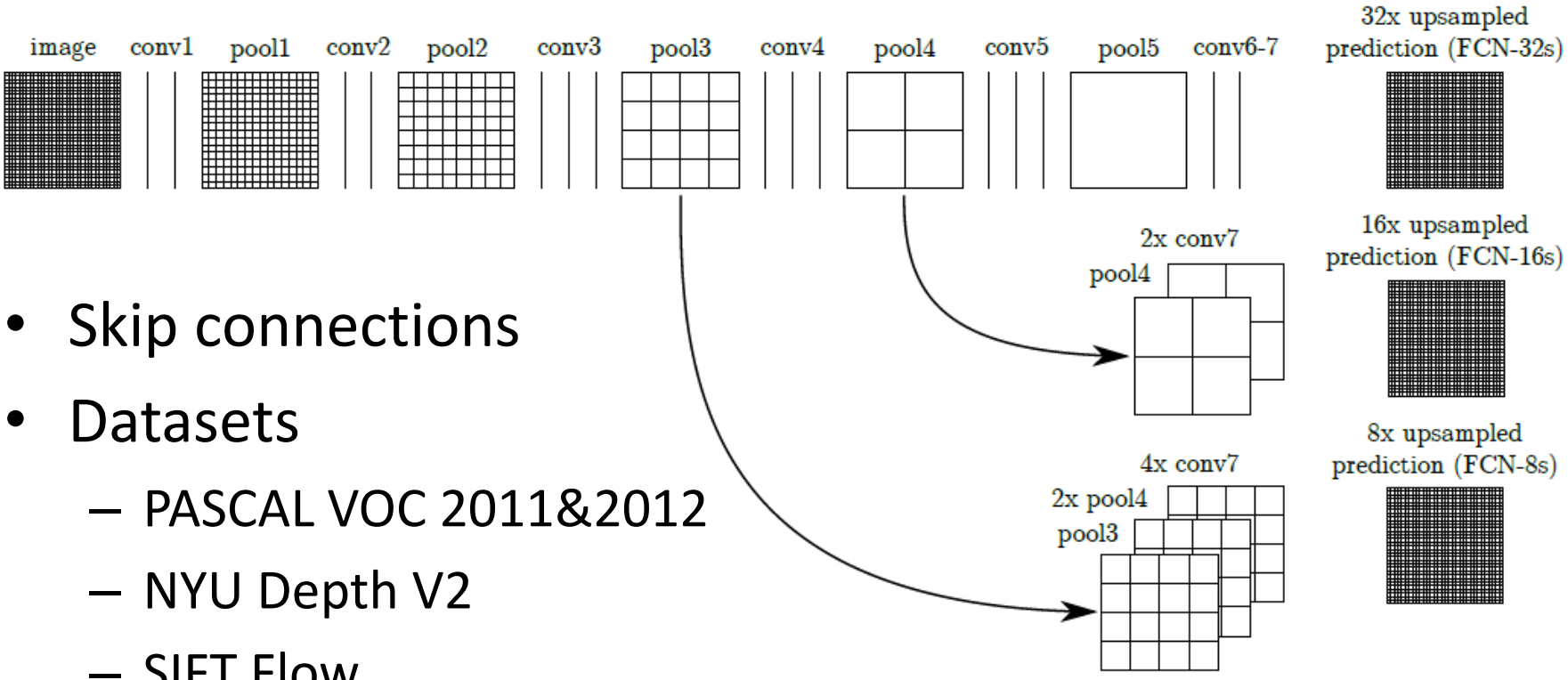
AlexNet
VGG-16
GoogLeNet



- Increasing the size of the feature maps
  - Bilinear interpolation
  - Transposed convolution

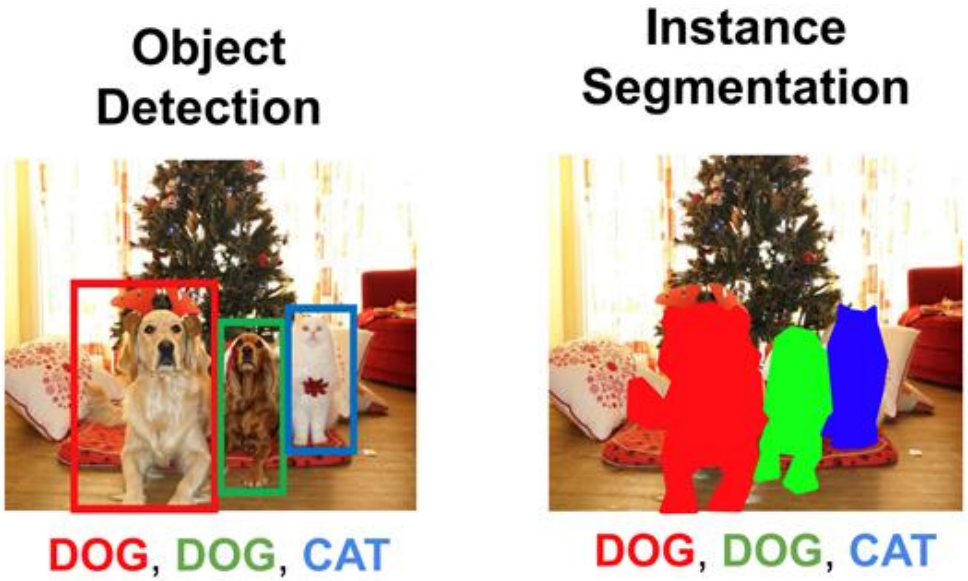
Fully Convolutional Networks

Long et al., CVPR 2015



- Skip connections
- Datasets
  - PASCAL VOC 2011&2012
  - NYU Depth V2
  - SIFT Flow

Instance segmentation

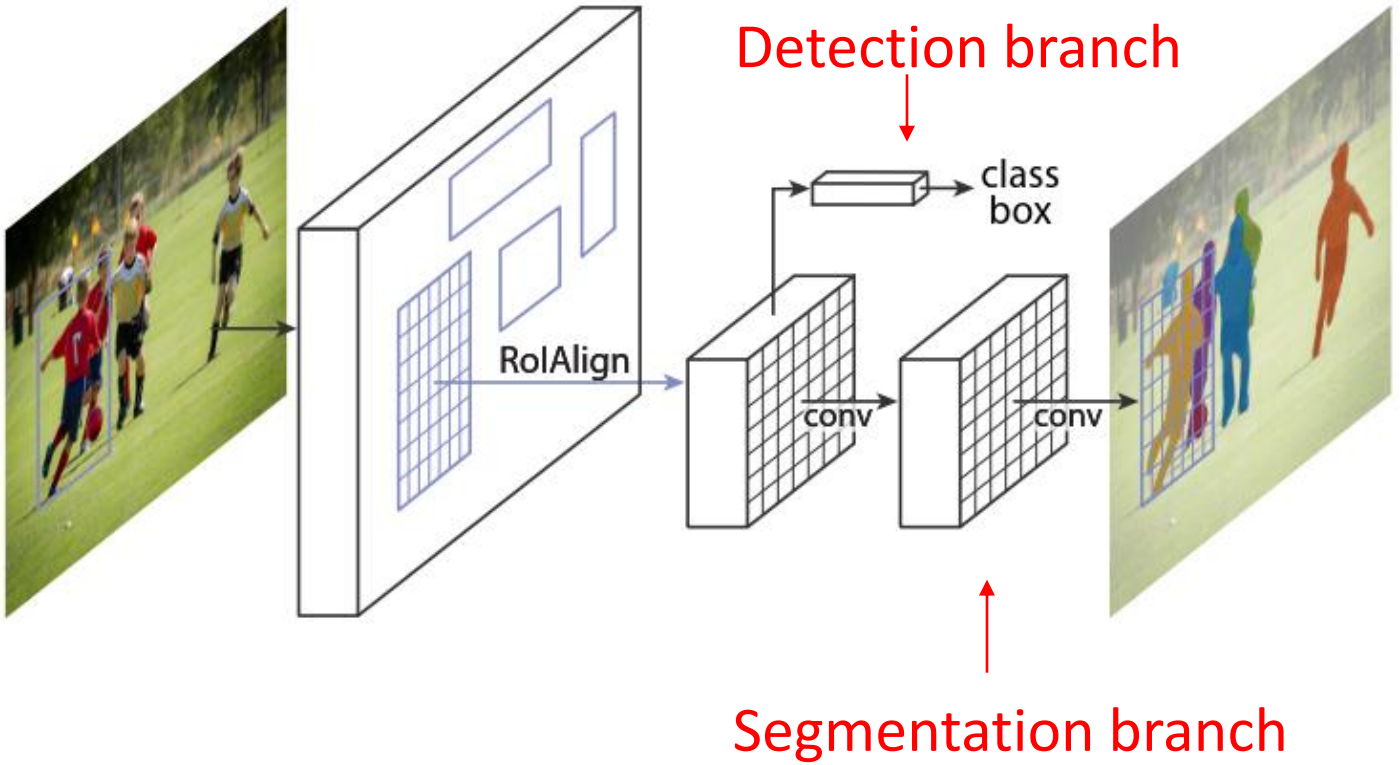


How would you solve this problem?

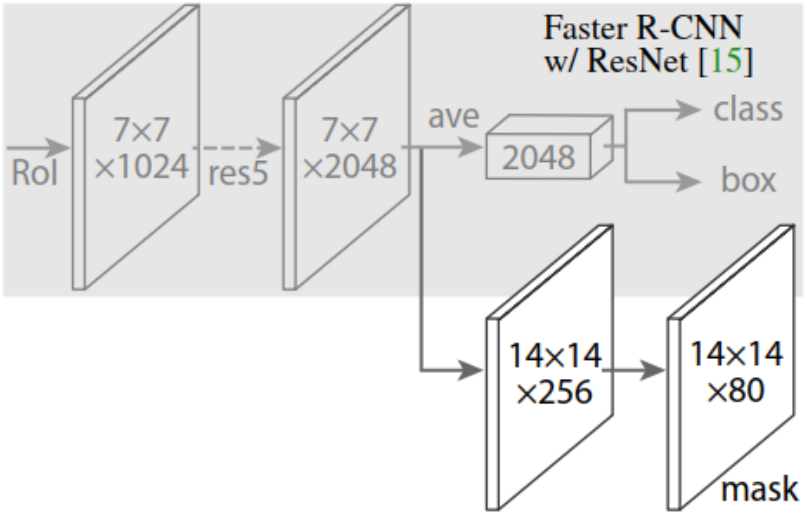
Mask R-CNN

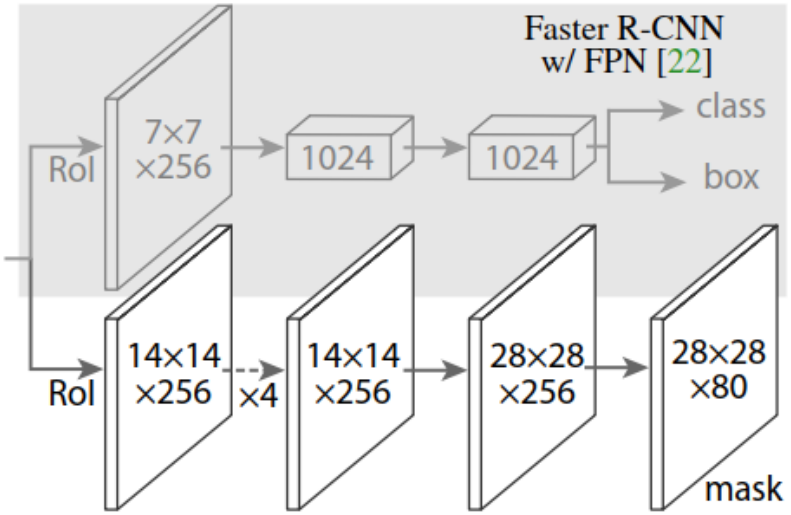
He et al., ICCV 2017

Simultaneous detection and segmentation



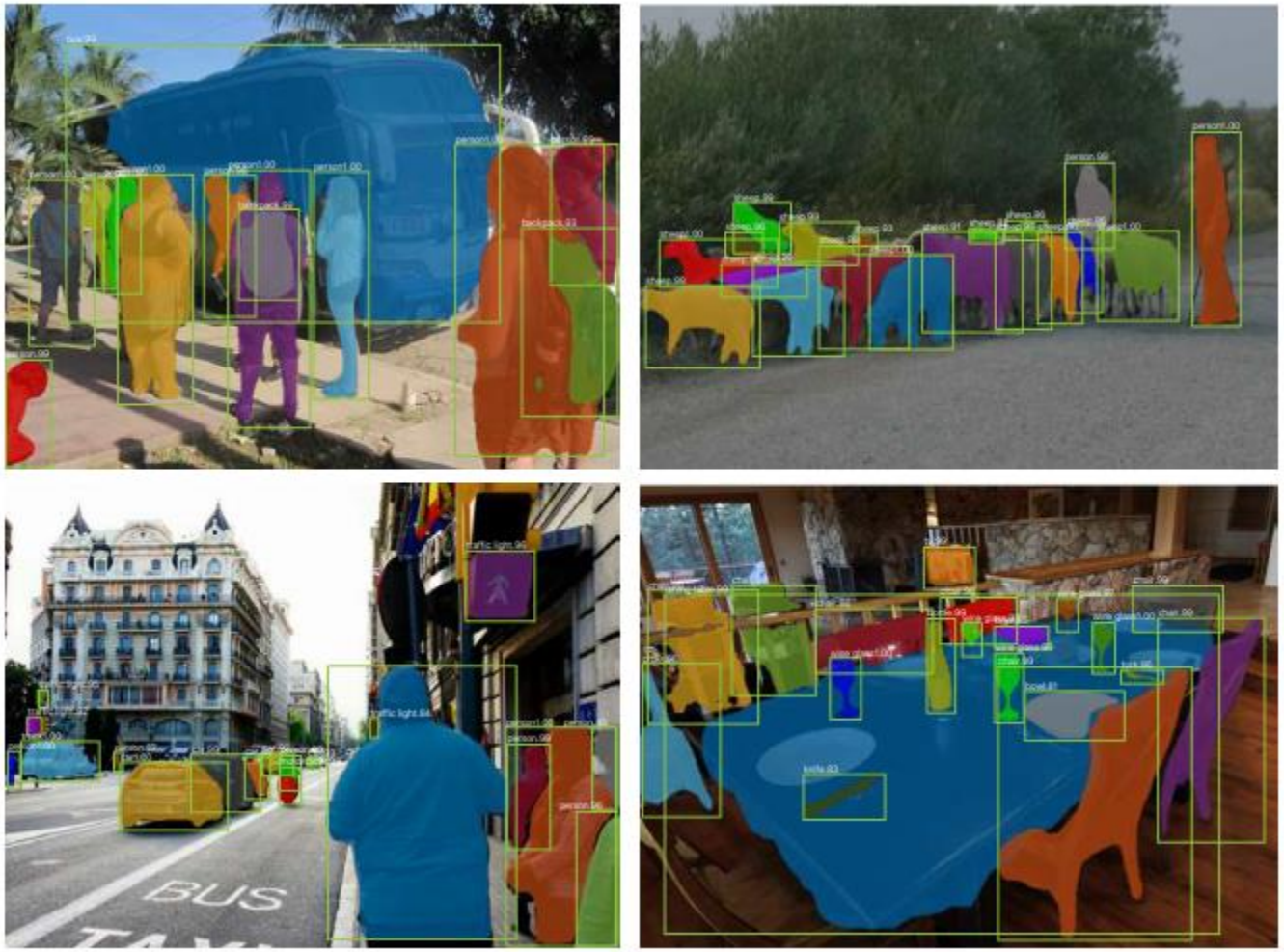
Head architecture





FPN: Feature pyramid network (Lin et al., 2017), an efficient backbone for object detection

Results



Summary of Part III

Image segmentation tasks

Semantic Segmentation



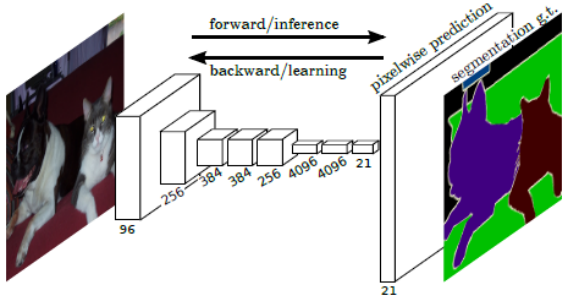
GRASS, CAT, TREE, SKY

Instance Segmentation

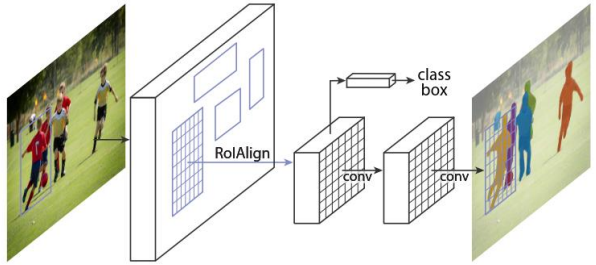


DOG, DOG, CAT

FCN



Mask R-CNN



Outline

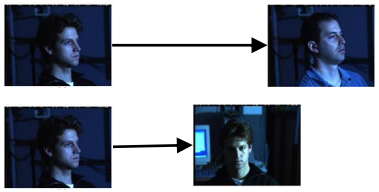
I. Image classification
II. Object detection
III. Image segmentation
IV. Summary

Summary of this lecture

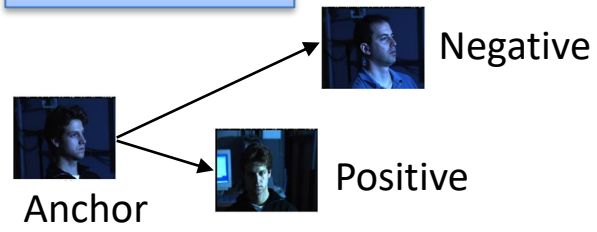
Knowledge

I. Image classification

Contrastive loss

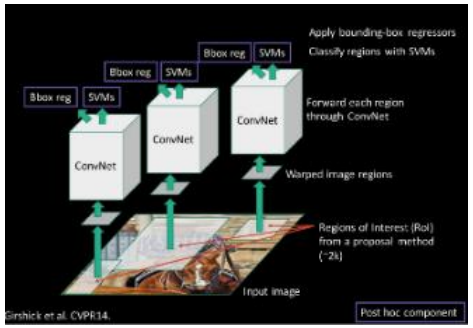


Triplet loss

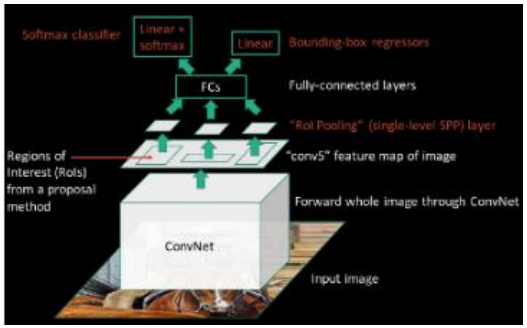


II. Object detection

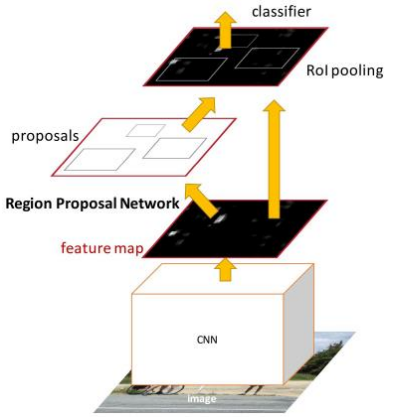
R-CNN



Fast R-CNN



Faster R-CNN

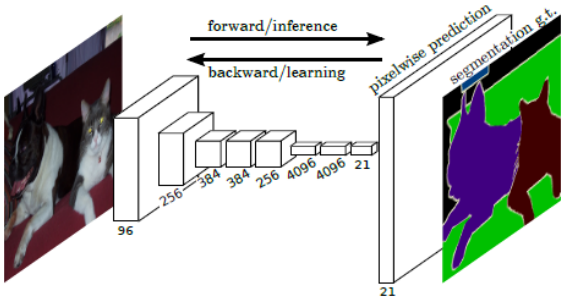


Summary of this lecture

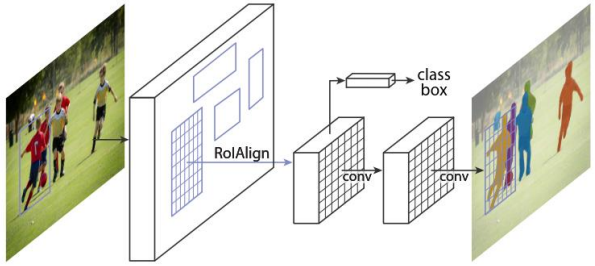
Knowledge

III. Image segmentation

FCN



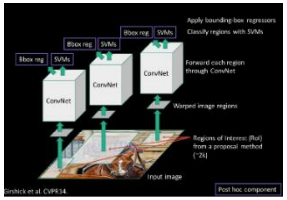
Mask R-CNN



Summary of this lecture

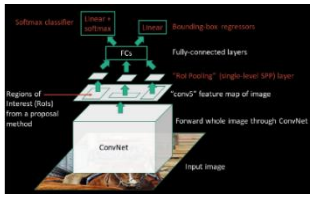
Capability and value

R-CNN



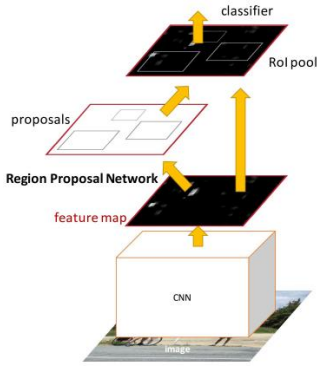
Too many feedforward passes

Fast R-CNN



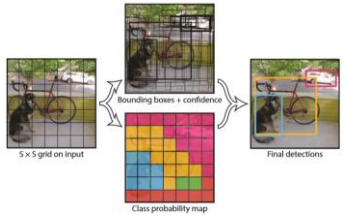
External models for RoI generation

Faster R-CNN



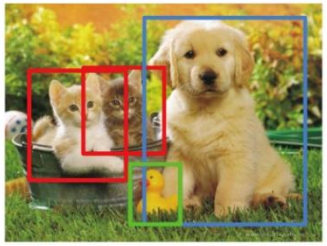
2 stages are slow

YOLO



Accuracy is low

Identify problems, solve problems



1. Predict lots of RoIs
2. Do classification for each RoI

Google scholar citations (till Nov 10, 2025):

45183

Simple ideas, influential works

Main references

- Sun, Wang, Tang (2014)
  Deep Learning Face Representation by Joint Identification-Verification
  NeurIPS
- Schroff, Kalenichenko, Philbin (2015)
  FaceNet: A Unified Embedding for Face Recognition and Clustering
  CVPR
- Girshick, Donahue, Darrell, Malik (2014)
  Rich feature hierarchies for accurate object detection and semantic segmentation
  CVPR
- Girshick (2015)
  Fast R-CNN
  ICCV

Main references

- Ren, He, Girshick, Sun (2015)

  Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks

  NeurIPS
- Long, Shelhamer, Darrell (2015)

  Fully Convolutional Networks for Semantic Segmentation

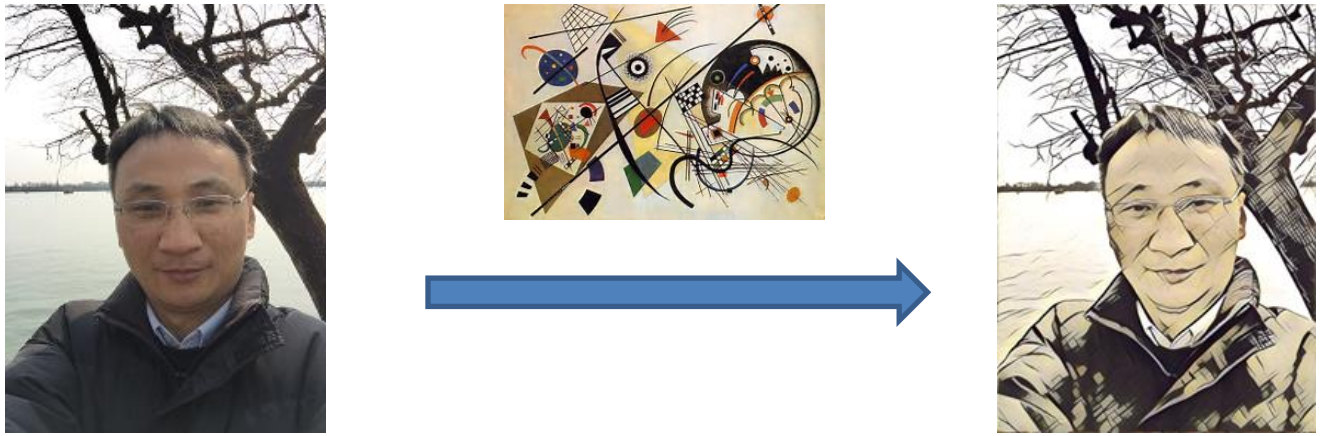
  CVPR
- He, Gkioxari, Dollar, Girshick (2016)

  Mask R-CNN

  ICCV

Recommended reading: Image style transformation

Gatys, Leon A., Alexander S. Ecker, and Matthias Bethge. "A neural algorithm of artistic style." CVPR 2016.



HwX3: Brain MRI image segmentation

Deadline: Nov 24 (Monday noon) 12:00

- Datasets:
  - Brain MRI (https://www.kaggle.com/mateuszbuda/lgg-mri-segmentation)
- Submit all codes and a report.
- The report should detail all improvement attempts and corresponding results (including DSC values and example results), whether or not they successfully improved the model's effectiveness and an analysis of the results.

